



GIRRAWEE HIGH SCHOOL

HALF-YEARLY EXAMINATION

YEAR 12

2015

MATHEMATICS

Time allowed – Two hours

(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt all questions.
- Circle the best response for the questions in Part A.
- Start each question in Part B on a new page.
- All necessary working must be shown.
- Marks may be deducted for careless or badly arranged work.

PART A (10 marks)

Question 1

0.0398 to 2 significant figures is:

- (a) 0.03 (b) 0.04 (c) 0.039 (d) 0.040

Question 2

The solution to $35 - 3x > 19 - x$ is:

- (a) $x < 8$ (b) $x > 8$ (c) $x \leq 8$ (d) $x \geq 8$

Question 3

If α and β are the roots to the equation $x^2 + 4x + 1 = 0$, then the value of $\alpha + \beta$ is:

- (a) $\frac{1}{4}$ (b) 4 (c) $-\frac{1}{4}$ (d) -4

Question 4

The perpendicular distance from the point (2, 3) and the line $6x + 8y - 5 = 0$ is:

- (a) $\frac{31}{10}$ (b) $\frac{31}{100}$ (c) $\frac{17}{10}$ (d) $\frac{17}{100}$

Question 5

Evaluate $\sum_{x=1}^4 x^x$

- (a) 1 (b) 4 (c) 256 (d) 288

Question 6

The focus of $x^2 = -16y$ is:

- (a) (0, 4) (b) (4, 0) (c) (0, -4) (d) (-4, 0)

Question 7

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

- (a) 0 (b) $\frac{1}{3}$ (c) 3 (d) 6

Question 8

The second derivative of $f(x) = x^2 + \frac{1}{x}$

- (a) $2x - \frac{2}{x^2}$ (b) $2 - \frac{1}{x^3}$ (c) $2 + \frac{2}{x^3}$ (d) $2x + \frac{2}{x^2}$

Question 9

The solution to $|2x - 1| \geq 3$

- (a) $x \geq 2$, $x \leq -1$ (b) $x \leq 2$, $x \geq -1$ (c) $x \geq -2$, $x \leq 1$ (d) $x \leq -2$, $x \geq 1$

Question 10

Which definite integral represents the area bounded by the curve $y = 4 - x^2$ and the x axis?

- (a) $\int_0^2 (4 - x^2) dx$ (b) $\int_{-2}^0 (4 - x^2) dx$ (c) $\int_{-2}^2 (4 - x^2) dx$ (d) $\int_{-4}^4 (4 - x^2) dx$

PART B

Question 11 (17 marks)

(a) Differentiate the following. Simplify the answer if necessary.

(i) $y = x^6 + 2\sqrt{x}$ (1)

(ii) $y = 2e^{4x} + e^{-x}$ (2)

(iii) $y = e^{2x} (e^x - e^{-x})$ (2)

(iv) $y = x^2 e^{x^3}$ (3)

(v) $y = \frac{e^{4x}}{x-1}$ (3)

(b) Simplify $\log_6(16) + \log_6(81)$ (3)

(c) Use the trapezoidal rule with 5 function values to give an

approximation for $\int_3^7 (x^2 + 3x)dx$. (3)

Question 12 (15 marks)

(a) $P(2, 3)$, $Q(6, -1)$ and $R(-4, -5)$ are the vertices of $\triangle PQR$. M is the midpoint of PQ and N is the midpoint of PR .

(i) Draw a diagram showing this information. (1)

(ii) Find the co-ordinates of M and N . (2)

(ii) Show that $MN \parallel QR$. (2)

(iii) Calculate the length of QR . (2)

(iv) Show that the length of MN is half the length of QR . (2)

(b) Evaluate $\int_0^1 e^{-\frac{x}{2}} dx$ using Simpson's rule with 3 function values, giving your answer as an exact value. (3)

(c) Solve for a , $\log_2 a - \log_2 3 = \log_2 (a+2) - \log_2 (a-2)$ (3)

Question 13 (16 marks)

- (a) Consider the curve given by $y = 1 + 3x - x^3$, for $-2 \leq x \leq 3$.
- (i) Find the stationary points and determine their nature. (4)
- (ii) Find the point of inflexion. (2)
- (iii) Sketch the curve for $-2 \leq x \leq 3$. (3)
- (iv) What is the minimum value of y for $-2 \leq x \leq 3$. (1)
- (b) A box contains 8 red and 11 green marbles. Arnav randomly selects three marbles one at a time and without replacement. What is the probability that he selects green, red then green in that order? (2)
- (c) In a class of 24 students, 3 play no sport, 14 play cricket and 12 play tennis.
- (i) Construct a Venn Diagram showing this information. (1)
- (ii) If a student is selected at random, find the probability that:
- (α) He or she plays tennis? (1)
- (β) He or she plays both tennis and cricket? (1)
- (γ) He or she doesn't play cricket? (1)

Examination continues on the next page

Question 14 (20 marks)

(a) Find the following integrals.

(i) $\int 5x^3 - 2x + \sqrt{x} \, dx$ (2)

(ii) $\int e^x + e^{-3x} \, dx$ (2)

(b) (i) Differentiate $y = e^{x^2}$ (2)

(ii) Hence, find $\int_0^2 2xe^{x^2} \, dx$ (2)

(c) Find the sum of the first twenty terms of an arithmetic series, given that the tenth term is 39 and the sum of the first ten terms is 165. (4)

(d) Find the values of x for which the series $1 + (x-3)^2 + (x-3)^4 + \dots$ has a limiting sum. (4)

(e) Cans of fruit in a supermarket display are stacked so that there are 3 cans in the top row, 5 in the next row, 7 in the next row and so on. If there are 10 rows in the display, find:

(i) The number of cans in the bottom row (2)

(ii) The total number of cans in the display (2)

Question 15 (15 marks)

(a) A company finds that the function $f(x) = x^3 - 96x^2 + 2880x$ provides a good approximation for their profit $f(x)$ in dollars, where x is the advertising expenditure in thousands of dollars.

(i) What expenditure of advertising would produce the maximum profit? (4)

(ii) What is the maximum profit? (2)

(b) Find the value of k if $y = e^{kx}$ is a solution of $y'' - y' - 12y = 0$. (4)

(c) Sketch the graphs $y = e^{-x}$, $y = x + 1$ and the line $x = 2$. Find the area of the region bounded by all three curves. (5)

Question 16 (14 marks)

- (a)** A piece of wire 1 m long is cut into two parts. Each part is bent to form a circle. If one of the pieces of wire is x cm long:

(i) Show that the length in centimetres of the radii of the two circles are

$$\frac{x}{2\pi} \text{ and } \frac{100-x}{2\pi}. \quad (2)$$

(ii) Show that the sum of the areas of the circles is

$$\frac{1}{2\pi}(x^2 - 100x + 5000) \text{ cm}^2. \quad (4)$$

(iii) Find the lengths of each piece of wire so that the sum of their areas is least. (4)

- (b)** A hemispherical bowl of radius a units is filled with water to a depth of $\frac{a}{2}$ units. Show that the volume of water is $\frac{5a^3\pi}{24}$ cubic units. (4)

END OF EXAMINATION 😊

YR 12 2U Half Yearly 2015.

PART A Multiple Choice.

1. D 2. A 3. D 4. A 5. D.
6. C 7. B 8. C 9. A 10. C.

PART B - Question 11

a) $y = x^6 + 2\sqrt{x}$

$y' = 6x^5 + \frac{1}{\sqrt{x}}$ (1)

ii) $y = e^{2x}(e^x - e^{-x})$
 $= e^{3x} - e^x$

$y' = 3e^{3x} - e^x$ (2)

v) $y = x^2 e^{x^3}$

$y' = e^{x^3}(2x) + x^2(3x^2 e^{x^3})$
 $= 2xe^{x^3} + 3x^4 e^{x^3}$
 $= xe^{x^3}(2 + 3x^3)$ (3)

vi) $y = \frac{e^{4x}}{x-1}$

$u = e^{4x}$
 $u' = 4e^{4x}$
 $v = x-1$
 $v' = 1$

$y' = \frac{(x-1)(4e^{4x}) - e^{4x}(1)}{(x-1)^2}$

$= \frac{4xe^{4x} - 4e^{4x} - e^{4x}}{(x-1)^2}$ (3)

$= \frac{e^{4x}(4x-5)}{(x-1)^2}$

b) $\log_6 16 + \log_6 81$

$= \log_6 2^4 + \log_6 3^4$

$= 4(\log_6 2 + \log_6 3)$ (3)

$= 4(\log_6 6)$

$= 4$

c) i) $\int_3^7 (x^2 + 3x) dx$

$h = \frac{7-3}{4} = 1$

	1	1	1	1	1
x	3	4	5	6	7
$f(x)$	18	28	40	54	70

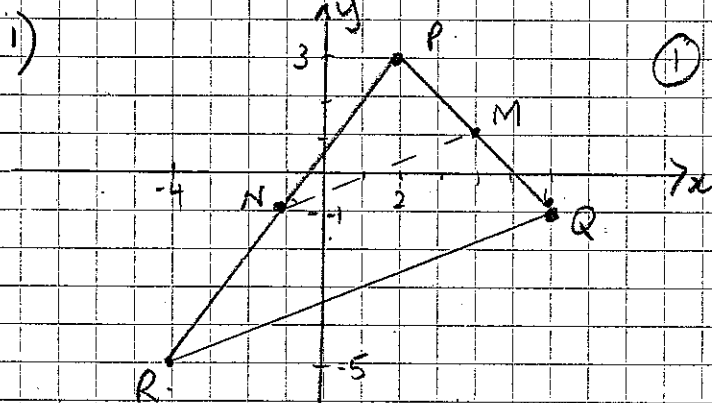
$\int_3^7 (x^2 + 3x) dx =$

$\frac{1}{2} [18 + 70 + 2(28 + 40 + 54)]$

$= \frac{332}{2} = 166$ (3)

Question 12

a) $P(2,3)$ $Q(6,-1)$ $R(-4,-5)$



ii) $M_{PQ} = \left(\frac{2+6}{2}, \frac{3+(-1)}{2}\right) = (4, 1)$

$M_{PR} = \left(\frac{2+(-4)}{2}, \frac{3+(-5)}{2}\right) = (-1, -1)$ (2)

$\therefore M(4, 1)$ $N(-1, -1)$

iii) $MN \parallel QR$

$m_{MN} = \frac{-1+1}{-1+4} = \frac{2}{5}$ $m_{QR} = \frac{-1+5}{6+4} = \frac{2}{5}$

$\therefore m_{MN} = m_{QR} \therefore MN \parallel QR$ (2)

iv) $d_{QR} = \sqrt{(6+4)^2 + (-1+5)^2}$
 $= \sqrt{10^2 + 4^2}$ (2)
 $= \sqrt{116}$

v) $MN^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$
 $= (4+1)^2 + (1+1)^2$

$= 5^2 + 2^2$ (2)

$MN^2 = 29$

$MN = \sqrt{29}$ units

Now $2 \times \sqrt{29} = \sqrt{4} \times \sqrt{29}$

$= \sqrt{116}$ units

$\therefore 2 \times MN = QR$

ie $MN = \frac{1}{2} QR$

b) $\int_0^1 e^{-\frac{x}{2}} dx$

$h = \frac{1-0}{2} = \frac{1}{2}$

x	0	$\frac{1}{2}$	1
$f(x)$	1	$\frac{1}{e^{\frac{1}{2}}}$	$\frac{1}{e}$

$\int_0^1 e^{-\frac{x}{2}} dx \hat{=}$

$\frac{1}{2} \left[1 + \frac{1}{\sqrt{e}} + 4 \left(\frac{1}{e^{\frac{1}{2}}} \right) \right]$

$= \frac{1}{6} \left(1 + \frac{1}{\sqrt{e}} + \frac{4}{e^{\frac{1}{2}}} \right)$ (3)

c) $\log_a - \log_3 = \log_2(a+2) - \log_2(a-2)$

$\log_2 \frac{a}{3} = \log_2 \frac{(a+2)}{(a-2)}$

$\therefore \frac{a}{3} = \frac{a+2}{a-2}$

$a(a-2) = 3(a+2)$ (3)

$a^2 - 2a = 3a + 6$

$a^2 - 5a - 6 = 0$

$(a-6)(a+1) = 0$

$\therefore a = 6$ $a = -1$

$\therefore a = 6$ as $a > 0$

Question 13.

a) $y = 1 + 3x - x^3$, $-2 \leq x \leq 3$.

i) $y' = 3 - 3x^2$ $y' = 0$.

$3 - 3x^2 = 0$.

$3x^2 = 3$.

$x^2 = 1$.

$\therefore x = \pm 1$.

When $x = 1$

$y = 3$.

When $x = -1$

$y = -1$.

$\therefore$ Stationary pts at $(1, 3)$ and

$(-1, -1)$

$y'' = -6x$.

When $x = 1$ $y'' = -6$ $y'' < 0$

$\therefore$ max.

When $x = -1$ $y'' = 6$ $y'' > 0$.

$\therefore$ min.

$(1, 3)$ is a max. turning pt.

$(-1, -1)$ is a min. turning pt.

ii) When $y'' = 0$.

$-6x = 0$.

When $x = 0$.

$y = 1$.

$\therefore x = 0$.

$\therefore$ possible point of inflexion

Test.

x	-1	0	1
y''	6	0	-6

change in

concavity

$\therefore (0, 1)$ is a point of inflexion.

iii) When $x = -2$

$x = 3$

$y = 1 + 3(-2) - (-2)^3$

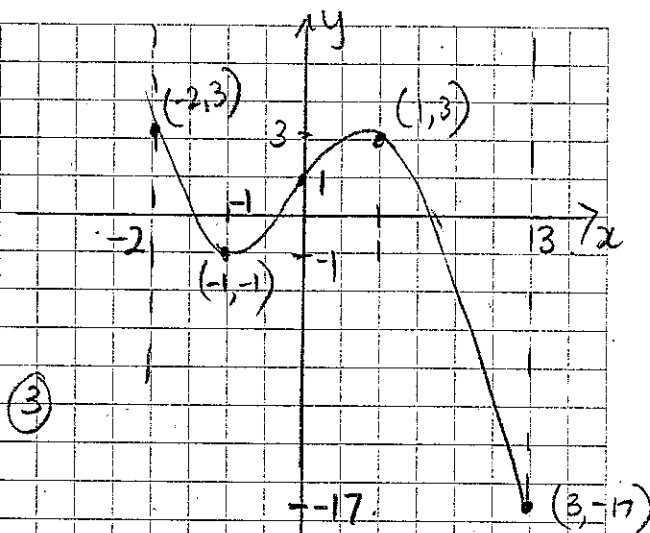
$= 1 - 6 + 8$

$= 3$

$y = 1 + 3(3) - (3)^3$

$= 1 + 9 - 27$

$= -17$



iv) minimum value y is -17 .

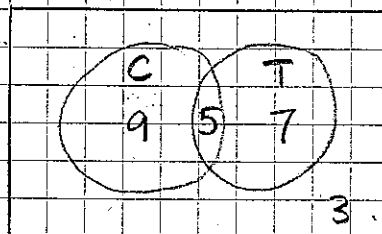
b) 8R 11G

$P(\text{GRG}) = \left(\frac{11}{19} \times \frac{8}{18} \times \frac{10}{17} \right)$

$= \frac{880}{5814}$

$= \frac{440}{2907}$

c) i)



ii) $P(\text{plays tennis}) = \frac{12}{24} = \frac{1}{2}$

iii) $P(\text{plays both}) = \frac{5}{24}$

iv) $P(\text{Don't play cricket})$

$= \frac{10}{24} = \frac{5}{12}$

Question 14.

a) i) $\int 5x^3 - 2x + \sqrt{x} \, dx$
 $= \frac{5x^4}{4} - x^2 + \frac{2}{3} \sqrt{x^3} + C$ (2)

ii) $\int e^x + e^{-3x} \, dx$
 $= e^x - \frac{1}{3} e^{-3x} + C$ (2)

b) i) $y = e^{x^2}$
 $y' = 2xe^{x^2}$ (1)

ii) $\int_0^2 2xe^{x^2} \, dx$
 $= \left[e^{x^2} \right]_0^2$
 $= e^4 - e^0$
 $= e^4 - 1$ (3)

c) $T_{10} = 39$ $S_{10} = 165$

$T_{10} = a + 9d$ $S_{10} = \frac{10}{2} [2a + 9d]$

$a + 9d = 39$ (1) $5(2a + 9d) = 165$

$2a + 9d = 33$ (2)

$a = -6$

$\therefore -6 + 9d = 39$

$9d = 45$ (4)

$d = 5$

$\therefore S_{20} = \frac{20}{2} [2(-6) + 19(5)]$

$= 10 [-12 + 95]$

$= 830$

d) $r = \frac{(x-3)^2}{(x-3)^2} = \frac{(x-3)^4}{(x-3)^2} = (x-3)^2$

$|r| < 1$ $\therefore$ A geometric series.

$(x-3)^2 < 1$

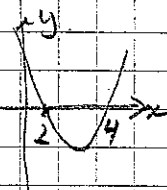
$x^2 - 6x + 9 < 1$ (4)

$x^2 - 6x + 8 < 0$

$(x-4)(x-2) < 0$

$x = 4, 2$

$\therefore 2 < x < 4$



e) $a = 3$ $d = 2$

i) $T_{10} = 3 + 9(2)$
 $= 3 + 18 = 21$ (2)

$\therefore$ There are 21 cans in the bottom row.

ii) $S_{10} = \frac{10}{2} [2(3) + 9(2)]$
 $= 5 [6 + 18]$
 $= 120$ (2)

$\therefore$ There are 120 cans on display.

Question 15.

$$a) f(x) = x^3 - 96x^2 + 2880x$$

$$i) f'(x) = 3x^2 - 192x + 2880$$

$$f'(x) = 0$$

$$3x^2 - 192x + 2880 = 0$$

$$x^2 - 64x + 960 = 0$$

$$(x - 40)(x - 24) = 0$$

(4)

$$\therefore x = 40, 24$$

$$f''(x) = 6x - 192$$

$$\text{When } x = 40 \quad f''(x) = 6(40) - 192 \\ = 240 - 192 \\ = 48$$

$$f''(x) > 0 \therefore \text{min.}$$

$$\text{When } x = 24 \quad f''(x) = 6(24) - 192 \\ = 144 - 192 \\ = -48$$

$$f''(x) < 0 \therefore \text{max.}$$

$\therefore$ \$24 would produce the maximum profit.

$$ii) f(24) = 24^3 - 96(24)^2 + 2880(24) \\ = 27648$$

(2)

$\therefore$ The maximum profit is

$$\$27648$$

$$b) y = e^{kx} \quad y'' - y' - 12y = 0$$

$$y' = ke^{kx}$$

$$y'' = k^2 e^{kx}$$

$$\therefore y'' - y' - 12y = k^2 e^{kx} - ke^{kx} - 12(e^{kx}) \\ = e^{kx}(k^2 - k - 12)$$

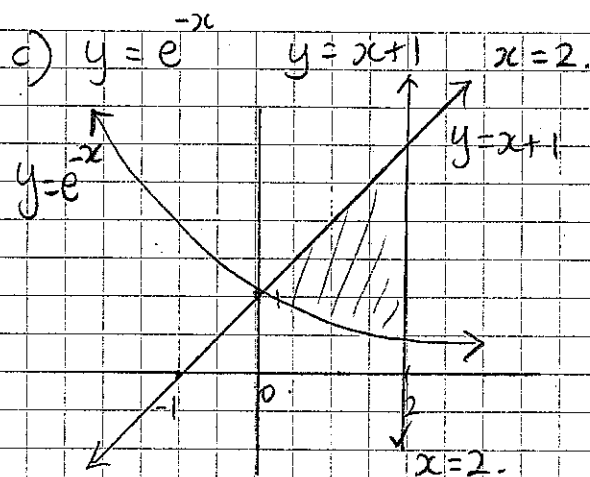
$$\therefore e^{kx}(k^2 - k - 12) = 0$$

$$k^2 - k - 12 = 0$$

$$(k - 4)(k + 3) = 0$$

(4)

$$\therefore k = 4, -3$$



$$A = \int_0^2 (x + 1) - e^{-x} dx \quad (5) \\ = \left[\frac{x^2}{2} + x + e^{-x} \right]_0^2 \\ = \left(\frac{2^2}{2} + 2 + e^{-2} \right) - \left(\frac{0^2}{2} + 0 + e^0 \right) \\ = \left(2 + 2 + \frac{1}{e^2} \right) - 1 \\ = 3 + \frac{1}{e^2} \text{ units}^2$$

Question 16

a) i) x and $100-x$ are the lengths of the two pieces of wire

Circle₁ $2\pi r = x$ Circle₂ (2)

$$\therefore r = \frac{x}{2\pi}$$

$$2\pi r = 100-x$$

$$\therefore r = \frac{100-x}{2\pi}$$

$$ii) A_1 = \pi r^2$$

$$A_2 = \pi r^2$$

$$= \pi \left(\frac{x}{2\pi} \right)^2$$

$$= \pi \left(\frac{100-x}{2\pi} \right)^2$$

$$= \frac{x^2}{4\pi} \text{ cm}^2$$

$$= \frac{(100-x)^2}{4\pi} \text{ cm}^2$$

$$A_1 + A_2 = \frac{x^2}{4\pi} + \frac{(100-x)^2}{4\pi}$$

(4)

$$= \frac{x^2 + (10000 - 200x + x^2)}{4\pi}$$

$$= \frac{2x^2 + 10000 - 200x}{4\pi}$$

$$= \frac{1}{2\pi} (x^2 - 100x + 5000) \text{ cm}^2$$

$$iii) \frac{d}{dx} \left[\frac{1}{2\pi} (x^2 - 100x + 5000) \right]$$

$$\frac{dA}{dx} = \frac{1}{2\pi} (2x - 100)$$

$$= \frac{1}{\pi} x - \frac{50}{\pi}$$

$$\text{When } \frac{dA}{dx} = 0$$

$$\frac{x}{\pi} - \frac{50}{\pi} = 0$$

$$x = 50$$

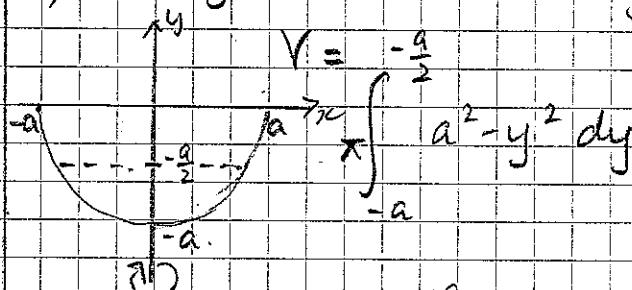
$$\frac{d^2A}{dx^2} = \frac{1}{\pi} \therefore \text{always positive}$$

$\therefore$ Only a minimum.

When $x = 50$ the sum of areas is least. (4)

$\therefore$ Each piece of wire is 50cm.

$$b) x^2 + y^2 = a^2 \therefore x^2 = a^2 - y^2$$



$$= \pi \left[a^2 y - \frac{y^3}{3} \right]_{-a}^a$$

$$= \pi \left[a^2 \left(\frac{a}{2} \right) - \left(\frac{a^3}{3} \right) \right]$$

$$- \left[a^2 \left(-a \right) - \left(\frac{-a^3}{3} \right) \right]$$

$$= \pi \left[\left(\frac{-a^3}{2} + \frac{a^3}{24} \right) - \left(-a^3 + \frac{a^3}{3} \right) \right]$$

$$= \pi \left[\left(\frac{-11a^3}{24} \right) - \left(\frac{-2a^3}{3} \right) \right]$$

(4)

$$= \pi \left[\frac{11a^3}{24} + \frac{16a^3}{24} \right]$$

$$= \frac{5a^3\pi}{24} \text{ units}^3$$